



REVIEW

Are microplastics actually harming our health?

REFERENCES 13 verified	STYLE ELI5	TOKENS ≈2.3k	VERIFICATION Crossref	COMPILED 27 Aug 2026
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SUMMARY

They may be harming us, and the warning signs are serious enough to study closely. But scientists still cannot say how much everyday exposure harms one person, or prove that plastic caused most of the illnesses seen so far.

Think about dust. You breathe and swallow tiny bits of the world around you without noticing. Microplastics are tiny pieces shed from plastic things. Some are smaller than a grain of sand, and some are far smaller. The big question is not only “Are they inside us?” It is “What do they do once they get there?”

01 Can plastic really get inside us?

Yes. Small studies have reported plastic material in blood, placentas, lungs, and brains. One study of 22 healthy people measured an average of 1.6 micrograms of plastic material in each millilitre of blood.¹ Another found 12 tiny fragments in 4 of 6 placentas.² A lung study found 39 pieces in 11 of 13 tissue samples.³ A later autopsy study found much more measured plastic in brain samples than in liver or kidney samples.⁴ The publisher later fixed scale bars and supporting-image details, while saying the spectra data used for the result did not change.⁵

That tells us exposure is real. It does not tell us that the plastic made anyone ill.

02 Have sick people been found to have more?

Sometimes. The clearest warning came from 257 people who needed surgery to clean a neck artery. Over about 34 months, heart attack, stroke, or death occurred in 20.0% of people whose removed artery gunk contained plastic, compared with 7.5% of those without detected plastic. The adjusted comparison was 4.53 times, with a plausible range from 2.00 to 10.27 times.⁶

Other studies found more plastic in stool from people with inflamed bowel disease and links between more kinds of plastic in semen and poorer sperm measurements.^{7,8} Across 25 studies in living people, these kinds of links appeared in several body systems. But most studies were small snapshots and had a moderate or high chance of mixing up cause with coincidence.⁹ A pregnancy review found only 7 studies covering 96 people, which was far too little to settle the question.¹⁰

You can see the evidence boundary in [Figure 1](#).

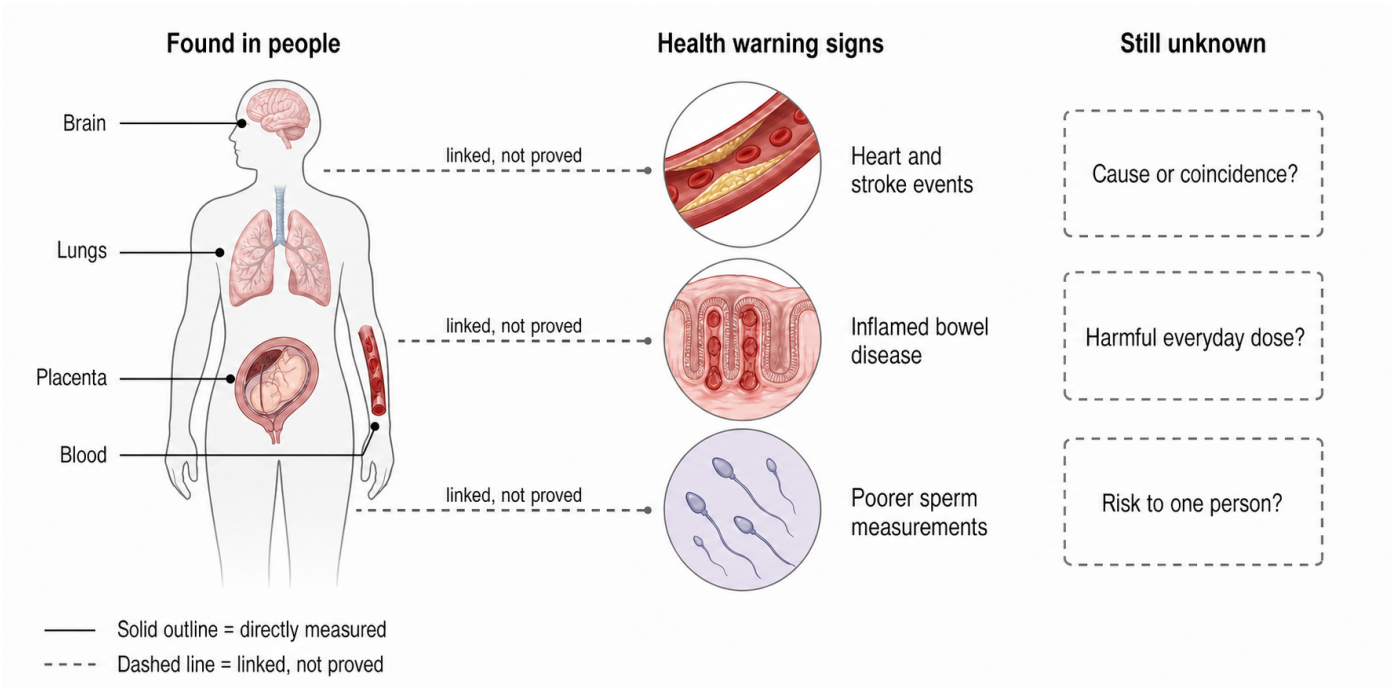


Figure 1. Finding plastic is not the same as proving harm. Human studies show that plastic material reaches several body parts. Some studies also find worrying health links. The dashed step means those links do not yet prove that plastic caused the illness, because dose, timing, other exposures, and measurement errors remain uncertain.^{9,6,11}

03 Why are scientists worried anyway?

In animal experiments, plastic particles can stir up inflammation, damage cells, and disrupt normal body signals. A careful review judged harm to the gut, lungs, and reproductive system to be suspected, not proven in people.¹² These experiments show that harm is possible. They do not reveal the danger from the much messier mix and dose that people meet in daily life.

04 But here's the thing: measuring tiny plastic is hard

Plastic is everywhere, including in lab air, clothing, tools, and containers. A sample can be polluted after it leaves the body. Different machines also see different particle sizes and report results in different units.¹³ In a 2026 check of 133 datasets from 101 human studies, not one met every essential quality rule for sampling, particle identification, and contamination control.¹¹

05 What is the honest answer?

Microplastics probably can harm living tissue, and early human findings are worrying. But we do not yet know the size of the health risk from ordinary daily exposure. Scientists need larger studies that follow healthy people for years, measure exposure the same way, and show that more exposure comes before more illness.

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